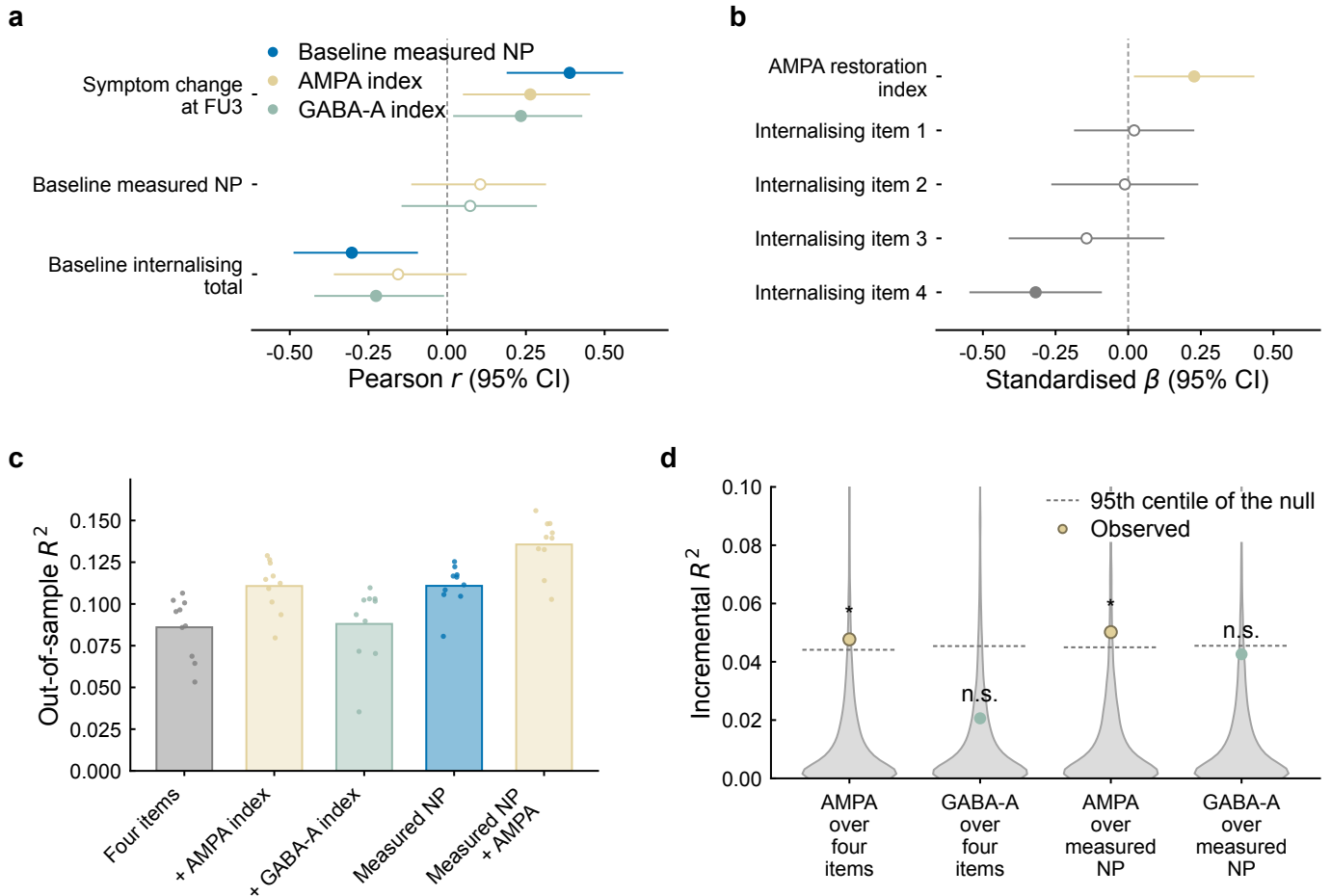




Supplementary Fig. S13 | The longitudinal prediction in full: what the restoration index adds over baseline behaviour and baseline connectivity. The follow-up sample is the $n = 85$ IMAGEN participants with a digital twin at baseline (age 19) and a symptom assessment at FU3 (age 23). The behavioural measure is the baseline internalising total, the sum of items 3-6 of the battery, and the outcome is that same sum at FU3 residualised on its baseline value; item-level correlations are tabulated in `long_correlations_n85.csv` and are not drawn here. The restoration index is the similarity of a participant's perturbed twin to the healthy pattern, computed separately for the AMPA and the GABA-A perturbation, with the same weights for both. Correlations are Pearson with Fisher-z 95% confidence intervals; all tests two-sided; filled symbols mark an uncorrected $P < 0.05$ in a and b. This figure holds the analyses behind Fig. 5h and 5i without repeating them: the index-versus-outcome scatter and the permuted R^2 of the full model are in the main figure. **a**, every pairwise correlation the prediction rests on. Measured baseline connectivity already carries signal: it tracks the baseline internalising total ($r = -0.30$ (95% CI -0.48 to -0.10 , $P = 0.00488$), Benjamini-Hochberg $q = 0.013$ across the panel's targets) and later symptom change ($r = +0.39$ (95% CI $+0.19$ to $+0.56$, $P = 0.00023$), $q = 0.002$), which is the baseline any claim for the twins has to beat. Both indices track later change (AMPA $r = +0.26$ (95% CI $+0.05$ to $+0.45$, $P = 0.0148$); GABA-A $r = +0.23$ (95% CI $+0.02$ to $+0.43$, $P = 0.0309$)) and neither is explained by the baseline scan (AMPA $r = +0.10$ (95% CI -0.11 to $+0.31$, $P = 0.34$); GABA-A $r = +0.07$ (95% CI -0.14 to $+0.28$, $P = 0.506$)), so an index is not a proxy for measured connectivity. Against the baseline internalising total the AMPA index is flat ($r = -0.16$ (95% CI -0.36 to $+0.06$, $P = 0.153$)) and the GABA-A index is weakly negative ($r = -0.23$ (95% CI -0.42 to -0.01 , $P = 0.0379$)). **b**, the full model of FU3 symptom change on the AMPA restoration index and the four internalising items: $R^2 = 0.233$, $F(5, 79) = 4.80$, $P = 0.0007$. The index carries an independent contribution (standardised beta = $+0.227$, 95% CI $+0.023$ to $+0.431$, $t(79) = 2.22$, $P = 0.0295$), as does one item (internalising item 4, beta = -0.319 , $P = 0.00603$); the other three do not ($P = 0.28$ - 0.93). The items enter separately here rather than as their sum, which is why they are listed individually; the item numbering follows the order of the battery and the per-column domain mapping is not verified in the source data. **c**, out-of-sample accuracy, 10-fold cross-validation with pooled out-of-fold predictions and 10 repeats (bars, mean over repeats; points, the individual repeats). Adding the AMPA index raises cross-validated R^2 from 0.086 ± 0.018 to 0.111 ± 0.016 over the four items, and from 0.111 ± 0.013 to 0.136 ± 0.016 over measured baseline connectivity; the GABA-A index adds nothing (0.088 ± 0.023). These are out-of-sample values and carry no test. **d**, incremental R^2 of each index over each covariate set, against its own permutation null (grey violin, 5,000 permutations of the outcome; dashed line, the null's 95th centile). Over the four items the AMPA index adds 0.048 ($F(1, 79) = 4.91$, $P = 0.0295$, permutation $P = 0.043$) and over measured baseline connectivity 0.050 ($F(1, 82) = 5.15$, $P = 0.0259$, permutation $P = 0.039$). The GABA-A index adds 0.021 ($P = 0.156$, permutation $P = 0.188$) and 0.043 ($P = 0.0405$, permutation $P = 0.059$), i.e. it reaches the conventional threshold over the measured scan but not over behaviour and not on the permutation test. The y axis is cut at 0.10 so that the observed values stay readable; the nulls are thin above that and reach 0.145 - 0.167 at their maxima, with 99th centiles of 0.074 - 0.078 . Symbols give the permutation P ; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; n.s., not significant. The increments are a few per cent of variance in a sample of 85, and the nulls are shown so that they can be read against what chance produces here.